Bass Model Analysis of Meta Ray-Ban Display

Marketing Analytics

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2025 Innovation: Meta Ray-Ban Display

Historical Look-alike Innovation: Smartphones

Scope: Worldwide

Most Relevant Historical Data: Worldwide Smartphone Sales by Year

Bass Model Parameters: p, q, M

1. Innovation Description

The innovation selected for this project is the Meta Ray-Ban Display, included in TIME's Best Inventions of 2025. The product combines traditional eyewear with a built-in digital display that allows users to view messages, calls, directions, photos, translations, and other information without taking out a smartphone. Meta pairs the glasses with the Meta Neural Band, an EMG wristband that translates subtle muscle signals and finger movements into commands for the glasses.

This product represents a new form of personal computing. Smartphones made digital information constantly accessible through a handheld device, while smart glasses attempt to make some of that information available directly in the user's field of view. TIME reports that the display is 600 by 600 pixels and that the Neural Band reads electrical signals from forearm muscles.

2. Historical Look-alike Innovation

The historical look-alike innovation selected for Meta Ray-Ban Display is the smartphone. Smartphones and smart glasses are both personal consumer electronic products that provide access to communication, information, and digital services. Smartphones allowed consumers to combine many functions into one portable device. Smart glasses attempt to provide some of those same functions in a wearable format.

Another similarity is that both products can be affected by social influence. Early smartphone users adopted the technology before it became common, while later users were more likely to adopt after seeing other people use smartphones successfully. Smart glasses may follow a similar pattern. Early demand may come mainly from technology enthusiasts, while broader adoption may depend on product improvements, reviews, social acceptance, and wider availability.

For these reasons, historical smartphone adoption provides a useful comparison for estimating the possible diffusion pattern of smart glasses.

3. Historical Data

The historical data used in this analysis are worldwide smartphone sales by year. The course material uses worldwide smartphone sales as an example of Bass Model estimation. The lecture reports sales of 122.32 million units in 2007, 139.29 million in 2008, 172.38 million in 2009, 296.65 million in 2010, 472.00 million in 2011, and 680.11 million in 2012. These data show how smartphone sales accelerated during the early growth period of the market.

4. Bass Model

The Bass Model explains new-product adoption using two main forces: innovation and imitation. The coefficient p represents the coefficient of innovation. It reflects adoption that occurs because of external influences such as advertising, media coverage, product information, and consumers' independent interest in a new product.

The coefficient q represents the coefficient of imitation. It reflects adoption caused by social influence, such as word of mouth, reviews, recommendations, ratings, and observing other consumers using the product. The third parameter, M, represents market potential, or the total number of potential adopters.

The Bass Model is expressed as: $f(t) / [1 - F(t)] = p + qF(t)$, where $f(t)$ is the fraction of the potential market that adopts during period t and $F(t)$ is the cumulative fraction of the market that has adopted by period t.

5. Estimation of Bass Model Parameters

The course lecture demonstrates estimation of the Bass Model parameters using worldwide smartphone sales and nonlinear least squares. The estimated values are p = 0.0053274, q = 0.3659114, and M = 17,449.76 in the same units used by the smartphone sales dataset. These estimates provide the historical diffusion reference for this project.

6. Interpretation of the Parameters

The estimated value of q is much larger than the estimated value of p. This suggests that imitation and social influence played a larger role in smartphone adoption than independent innovation. Early smartphone buyers may have been influenced by advertising, product announcements, and media attention. As smartphone ownership increased, later consumers were more likely to adopt because they saw other people using the technology and understood its benefits.

A similar process may occur with smart glasses. Early buyers may be attracted by novelty and technology, while later adoption may depend more heavily on social acceptance, product reviews, and visible use by other consumers.

7. Geographic Scope

The scope of this analysis is worldwide. A worldwide scope is appropriate because Meta Ray-Ban Display is a consumer technology intended for multiple international markets. Meta launched the product in the United States and announced expansion to Canada, France, Italy, and the United Kingdom. Using a worldwide scope also makes the comparison with global smartphone sales more consistent.

8. Applying the Smartphone Comparison to Meta Ray-Ban Display

The course material suggests that when forecasting a new product, a researcher can identify a similar historical product, estimate p and q from that product, and develop a separate estimate of M for the new product. For this analysis, the smartphone estimates p = 0.0053274 and q = 0.3659114 are used as historical diffusion parameters. They provide a starting point for modeling how Meta Ray-Ban Display and similar display-enabled smart glasses could diffuse through the market.

However, the market potential for smart glasses should not be assumed to be the same as the market potential for smartphones.

9. Market Potential Based on Published Forecast Data

Instead of using an unsupported assumption such as 100 million users, this analysis anchors the market potential discussion in published IDC forecast data. IDC forecasts optical see-through display glasses to grow from approximately 3.0 million units in 2026 to 12.2 million units in 2030, a compound annual growth rate of 41.9%.

IDC reports annual shipment forecasts rather than a lifetime number of unique adopters. Therefore, those shipment figures are not treated as a directly observed value of M. For a transparent modeling proxy, this report uses the cumulative 2026-2030 shipment volume implied by IDC's published 2026 level, 2030 level, and 41.9% CAGR. This produces a five-year cumulative shipment benchmark of about 34.0 million units. The value should be interpreted as a model input proxy, not as a measured lifetime market potential.

Year	IDC-implied OST display-glasses shipments (million units)
2026	3.00
2027	4.26
2028	6.04
2029	8.57
2030	12.16

Five-year cumulative shipment benchmark: 34.03 million units. The 2030 implied value rounds to 12.16 million, consistent with IDC's published 12.2 million figure.

10. Expected Diffusion Pattern and Adopters by Period

The Bass Model predicts that adoption is relatively slow in the early stages of a product's lifecycle. At first, most buyers are innovators who adopt because of product information, advertising, or personal interest. As more consumers begin using the product, the imitation effect becomes stronger. Potential buyers may see other people wearing smart glasses, read reviews, and become more comfortable with the technology.

Eventually, the number of new adopters reaches a peak. After that point, the number of new adopters begins to decline because a larger portion of the potential market has already adopted the product. The annual adoption curve is therefore typically bell-shaped, while cumulative adoption follows an S-shaped curve.

The table below applies the smartphone-based p and q parameters to the 34.0 million-unit market-potential proxy described above. These values are model estimates, not observed sales.

Year after introduction	Estimated new adopters (millions)	Estimated cumulative adopters (millions)
1	0.26	0.22
2	0.37	0.53
3	0.52	0.97
4	0.73	1.59
5	1.00	2.45
6	1.34	3.61
7	1.75	5.16
8	2.21	7.14
9	2.65	9.57
10	3.00	12.40
11	3.19	15.51
12	3.16	18.71
13	2.94	21.77
14	2.56	24.53
15	2.11	26.86

11. Estimated Peak Adoption

The Bass Model estimates the time of peak new adoption using $t^* = \ln(q/p) / (p + q)$. With $p = 0.0053274$ and $q = 0.3659114$, the estimated peak occurs at approximately 11.4 years after introduction. This does not mean adoption stops at that point. It means the number of new adopters is expected to be highest around that period and then gradually decline.

12. Limitations

There are several limitations to this analysis. First, smartphones and smart glasses are not identical products. Smartphones became essential communication devices, while smart glasses remain a more specialized wearable technology.

Second, the Bass Model parameters were estimated using historical smartphone sales. Smart-glasses customers may behave differently. Third, IDC's figures are shipment forecasts, not observed unique-adopter counts, so the 34.0 million value used here is a transparent modeling proxy rather than a directly measured lifetime market potential.

Finally, the basic Bass Model does not directly include variables such as price, competition, economic conditions, regulation, privacy concerns, product quality, or marketing expenditure. For these reasons, the results should be interpreted as a model-based scenario rather than an exact prediction.

13. Conclusion

Meta Ray-Ban Display represents a new step in wearable personal computing. The product combines eyewear, digital display technology, artificial intelligence, and gesture-based control through the Meta Neural Band.

This analysis uses smartphones as the historical look-alike innovation because both products provide consumers with personal access to digital information and communication. The Bass Model estimates derived from smartphone data show that imitation and social influence were much stronger than independent innovation during smartphone diffusion.

Applying the same logic to Meta Ray-Ban Display suggests that adoption may begin slowly and accelerate as the technology becomes more familiar, socially accepted, and widely available. Current IDC forecasts also support the view that display-enabled smart glasses are a rapidly growing segment. The Bass Model therefore provides a useful framework for analyzing the possible worldwide diffusion of Meta Ray-Ban Display, while recognizing that the forecast remains uncertain and depends on future market conditions.

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